Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/08/2011
PI Name:	Olabisi, Ronke Mojoyinola Ph.D.		
Project Title:	Bone formation through Biomineralization and Bioengineering		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI's current affiliation is City of Hope, Departments of Molecular Medicine and Surgery, Duarte, CA 91009. (Ed., September 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-03 Postdoctoral Fellowships
Start Date:	11/01/2008	**End Date:**	04/30/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	West, Jennifer (MENTOR/Rice University)		
Grant/Contract No.:	NCC 9-58-PF01902		
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	POSTDOCTORAL FELLOWSHIP Introduction: Understanding the signals that direct biomineralization is paramount for successful synthesis of self-assembled artificial bone and dentin biomimetic materials. Bone and seashell are biominerals shown to undergo both whole cell and single protein directed biomineralization. Furthermore, mammalian osteoblasts have been shown to produce shell-like structures in the presence of shells. Bone and nacre chips with no growth factors incubated with osteoblasts demonstrated biomineralization in the intervening gap: new bone near the bone chip and new nacre near the nacre chip. Understanding the concomitant changes in the microenvironment and their impact on osteoblast function and differentiation will enable the design of biomaterials that can create or direct specific functions. The hypothesis of this project is that water soluble proteins from seashell mother of pearl (nacre) cause osteoblasts to build bone and		

Task Description:	nacre. Nacre-initiated osteogenesis occurs in vivo, but not the production of new nacre. Despite its osteogenicity, nacre is not widely used as an implant material due to its limited dimensions and in vivo results that vary with preparation procedures. Regardless, nacre's presence changes the cellular microenvironment and causes functional changes in mammalian osteoblasts that direct them to build mollusk shell material. This behavior does not occur with the geological or synthetic mineral comprising nacre (aragonite), supporting our hypothesis that soluble factors are responsible. Nacre provides a model for engineering such factors into biomaterials. 1. SPECIFIC AIMS: 1) to determine the differences in gene expression of osteoblasts producing bone and osteoblasts producing nacre; 2) to determine the different proteins in the microenvironment of osteoblasts producing nacre vs. those producing bone; and 3) to pattern hydrogels with DMP1 and N40 (Dentin Matrix Protein 1, a hydroxyapatite nucleation protein in bone and dentin; and Nacre 40, an aragonite nucleation protein in nacre) to direct biomineralization in desired patterns. 2. KEY FINDINGS: 1) When osteoblasts are in the presence of nacre, they are induced to form bone more rapidly than without nacre present. 2) When osteoblasts are in the presence of nacre, they will produce altered mineralization, including mineralization that looks like nacre and mineralization that looks like highly rectangular bone formation, rather than the typical spheroid bone nodules. Yet when osteoblasts form this altered mineralization due to the presence of nacre, the effect is spontaneous rather than predictable. Out of 100 experiments, the altered mineralization occurred only once. 3) In the 99 experiments where there was no altered mineralization, there was a protein at 868.368 Da. When there was altered mineralization, this protein peak was absent. 4) Nacre proteins can be patterned into hydrogels in a specific pattern. 5) Osteoblasts will lay down bone on patterned hydrogels in the pattern defined. 3. IMPACT OF KEY FINDINGS The fact that the "nacregenesis" by osteoblasts was spontaneous rather than repeatable resulted in too few samples to perform gene analysis (specific aim 1). The protein identified at 868.368 Da that is absent when altered mineralization occurs may be a critical regulatory protein (specific aim 2). In future experiments, antibodies to this protein could be constructed to determine whether altered mineralization would occur with an antibody to the protein. This work was the first to ever demonstrate the capacity to pattern bone formation (specific aim 3). This is critical because many orthopedic biomaterials and agents can induce bone formation, but none have spatial control over where that bone will form. This is the first work that may provide such control, which is crucial in bone repair near the brain or spine. 4. PROPOSED RESEARCH PLAN FOR THE COMING YEAR: Not applicable, grant end.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Over 600,000 bone grafting procedures are performed annually in the United States alone, over 2.2 million worldwide. The autologous graft is the current clinical standard of care due to its ability to initiate bone formation and the fact that it poses no risk of rejection; however, it has drawbacks such as donor site morbidity and a limited supply. Bone morphogenetic proteins have been used increasingly to augment bone repair, but when used in certain locations complications such as swelling and bone formation in unwanted sites have been reported. Therefore, there remains an overwhelming need for alternatives to the autograft that are still capable of inducing bone formation in a controlled manner. Although bone formation is a type of biomineralization, there is very little crossover between the fields of bone biology and biomineralization. The discovery of the first successful endosseous implant using mother of pearl (the lustrous inner shell, nacre) in a 700 BC Mayan skull led to research into the effects of nacre on bone formation. To date, both whole nacre and its extracted proteins have demonstrated a capacity to elicit bone formation from osteoblasts both in vitro and in vivo. Concurrently, biomineralization researchers isolated nacre proteins responsible for exerting spatial control over the nucleation of aragonite, the calcium carbonate polymorph that comprises the mineral component of nacre. Recent findings suggest that the proteins directing seashell biomineralization are the same as those inducing bone formation. In addition to its potential as an alternative to the autograft, this biodegradable system inspired by seashells will be designed to provide spatial control over bone formation, which is crucial for orthopedic tissue engineering because spatial control allows for the design of replacement delicate bony structures such as the stapes in the ear and prevents unchecked bone formation, which can lead to complications in locations such as the skull or spine.
	During this year, I moved from using whole nacre pieces to extracting nacre proteins. This required removing the nacre from the prismatic layer, grinding the nacre to a fine powder and extracting the water soluble, urea soluble and EDTA soluble proteins. This was accomplished by combining the ground nacre powder with each respective solution, successively, and lyophilizing the supernatant from each. The resulting proteins were pegylated by reacting them with poly(ethylene glycol)-succinimidyl valerate (PEG-SVA) to attach crosslinkable acrylated PEG polymers. The pegylation of these proteins was confirmed by ninhydrin. PEG hydrogels were formed containing PEG-RGDS to enable osteoblast attachment. PEGylated nacre proteins were combined with fluorescein o-acrylate (a fluorescent marker) and placed on the surface of hydrogels in all combinations of the 3 proteins: 1. No proteins 2. Only fluorescent marker (FM) 3. Water soluble matrix (WSM) proteins+ FM 4. Urea soluble matrix (USM) proteins + FM 5. EDTA soluble matrix (ESM) proteins + FM 6. WSM+USM+FM

Task Progress:	7. WSM+ESM+FM 8. USM+ESM+FM These experiments were conducted by polymerizing the above 8 conditions over the whole hydrogel surface, or in specified patterns. To make patterns, a photomask was placed over the proteins and exposed to collimated white light in order to crosslink the PEGylated proteins to the hydrogel surface in the pattern. Murine osteoblasts (MC3T3-E1) were seeded on the surfaces of these hydrogels and bone formation was allowed to progress. The hydrogels were then analyzed for differences, which were notable. To ensure that the differences weren't due to steric hindrance of the nacre proteins compared to the RGDS, fibronectin was pegylated, polymerized into a bulk hydrogel and the experiments were repeated. On day 3, seeded hydrogels were imaged with phase and fluorescent microscopy and bone nodules were quantitated. On day 12, hydrogels were stained using the von Kossa method and imaged with phase microscopy. By day 3, mineralization occurred at a significantly greater rate (p=0.019) on the WSM hydrogel within or directly adjacent to areas containing WSM than areas devoid of the protein, as visualized by fluorescence of collocated fluorescein o-acrylate. Bone formation appeared at an increased rate in all the cases with WSM proteins, and localized within the patterns in the WSM-only hydrogels, as visualized with von Kossa stains. While WSM proteins have been demonstrated to initiate bone formation by MC3T3-E1 cells, this is the first time this capacity has been demonstrated when the proteins have been covalently immobilized on a patterned hydrogel. These results were the first to demonstrate microscopic spatial patterning of bone and were submitted to the American Society of Bone and Mineral Research and the Biomedical Engineering Society. The results were written up for publication on April 21st, and submission is currently awaiting approval from my mentor.
Bibliography Type:	Description: (Last Updated: 09/06/2019)
Abstracts for Journals and Proceedings	Olabisi RM, Lazard ZW, Olmsted-Davis EA, Davis AR, West JL. "The Effect of Nacre on Murine Osteoblasts in Culture." 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. Abstract #1131. , Feb-2010
Abstracts for Journals and Proceedings	Olabisi RM, Lazard ZW, Olmsted-Davis EA, Davis AR, West JL. "The Effect of Pegylated Nacre on Murine Osteoblasts in Culture." 18th International Academy of Astronautics Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th International Academy of Astronautics Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Olabisi RM, Rowland ML, Franco CS, Hoffmann, JC, West JL. "Forming Bone in Artificial Patterns Using Mother of Pearl." 2011 American Society of Bone and Mineral Research Annual Meeting, San Diego, CA, September 16-20, 2011. 2011 American Society of Bone and Mineral Research Annual Meeting, San Diego, CA, September 16-20, 2011. Presentation # SA0225. , Sep-2011
Abstracts for Journals and Proceedings	Olabisi RM, Rowland ML, Franco CS, Hoffmann, JC, West JL. "Patterning bone formation with mother of pearl." 2011 Biomedical Engineering Society Annual Meeting, Hartford, CT, October 12-15, 2011. 2011 Biomedical Engineering Society Annual Meeting, Hartford, CT, October 12-15, 2011. , Oct-2011
Articles in Other Journals or Periodicals	Olabisi RM, Rowland ML, Franco CS, Hoffmann, JC, West JL. "Patterning bone formation using mother of pearl." Nature Methods. Submitted, 2011. (As of September 2011.) (Ed. note: not yet published as of September 2019) , Sep-2011
Awards	Olabisi RM. "Outstanding Poster and Research Presentation Award from the Center for Biological and Environmental Nanotechnology Student Leadership Council, September 2009." Sep-2009
Awards	Olabisi RM. "The Gordon Research Conference's Carl Storm Underrepresented Minority Fellowship, March 2009." Mar-2009